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Assignment -03

1. Enumerate the failures of CFT and discuss the assumptions of quantum free e⁻s theory of metals.

⇒ The CFT would successfully Explain the Electrical & thermal Conductivity in metals.

- Specefic heat of metals: The molar heat Capacity of free e⁻s in a metal is given by 3/2 R. but Experimently the molar heat Capacity of metals is 10^{-4} RT which is Comparatively less value than the theoritically predicted value & also the dependency of temp T is not discussed.
- Dependance of Electrical Conductivity on temp:

$$1/2\ mv^2_{th} = 3/2\ KT \rightarrow (1)$$

where, $v^2_{th} = T$ or $v_{th} = \sqrt{T}$ since the mean Collision time is inversly proportional to the thermal velocity we write $\tau \propto 1/v_{th}$

but from the Eqⁿ of Conductivity we know

$$\sigma = \frac{ne^2\tau}{m}$$

- where as Experimentally it is observed that $\sigma \propto 1/T$ thus classical FT fails to Explain the temp dependence of Conductivity.

→ Energy levels / values of the Conductivity e⁻s are quantized.

- Distribution of e⁻s in various allowed levels taken place according to pauli Exclusion Principle
- Conduction e⁻s travel freely in a Const potential

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inside the metal, but stay confined with in the boundaries.

* Attraction blw the free e⁻s & the lattic ion or repulsion blw e⁻s themselves is ignored.

2. Define Fermi Energy level. Discuss the varies Energy states by the e⁻s at T=0K & T>0K on the basis of fermi factor.

⇒ Energy from fermi level defines which occupied highest level Energy from absolute 0°K is known as fermi Energy & Energy level is called fermi Energy level.

→ Probability of occupation for $E < E_F$ at T=0K: for Energy levels E, lying below E_F, $E - E_F$ taken a negative value.

$$f(E) = \frac{1}{1 + e^{-\infty}}$$

$$f(E) = 1$$

This implies that all Energy level below E_F are completely filled.

→ Probability of occupation for $E > E_F$ at T=0K: For Energy levels E, lying below E_F, $E - E_F$ taken a +ve values.

$$f(E) = \frac{1}{1 + e^{\infty}}$$

$$f(E) = 0$$

This implies that all Energy levels below E_F are completely vacant.

→ The probability of Occupation at ordinary temp $E = E_F$ at T>0K.

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At room temp, the probability starts reducing from 1 for values of E close to E_F, but larger than E_F

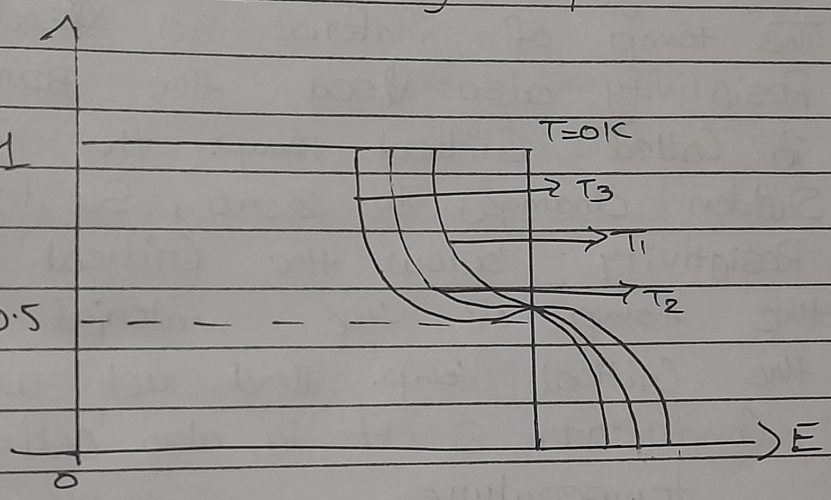
> At T≥0K, for $E = E_F$ the argument in the Exponential fun^n becomes zero.

$$f(E) = 1/1 + e^0$$

$$f(E) = 1/1+1$$

$$f(E) = 1/2 = 0.5$$

This implies that probability of occupancy of an e^- of 50% at any temp above 0K.



3. Calculate the probability of occupation of an Energy level 0.2eV above fermi level at temp 27°C

⇒ $E = E_F + 0.2eV$

$$E - E_F = 0.2 \times 1.6 \times 10^{-19}$$

$$E - E_F = 3.2 \times 10^{-20}$$

w.KT: $T = 27°C \Rightarrow 27 + 273 = 300$

$$f(E) = \frac{1}{1 + e^{\frac{E-E_F}{KT}}}$$

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$$= \frac{1}{1+e^{\frac{3.2\times10^{-20}}{1.38\times10^{-23}\times300}}}$$

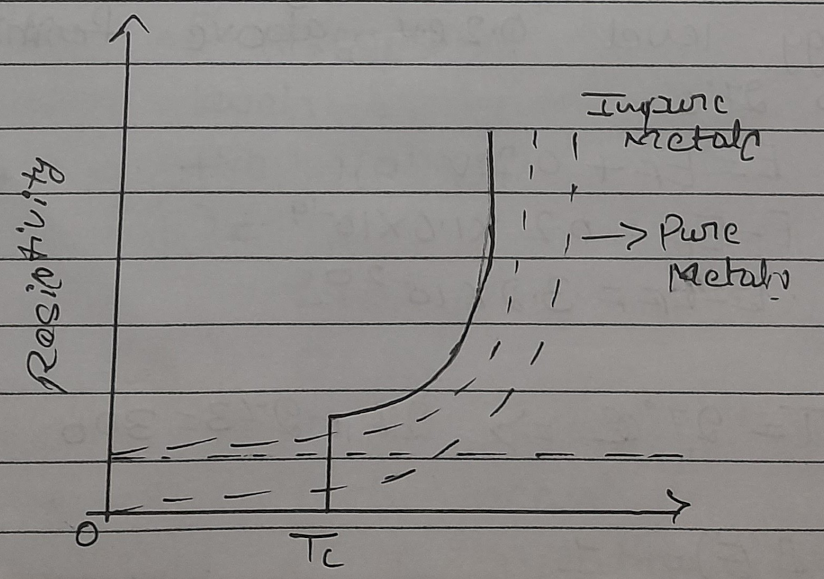
$$f(E) = 4.394\times10^{-4}$$

8. Describe Meissner Effect.

⇒ The material of Superconductor it completely blocks magnetic fields from entering inside. This makes it perfectly diamagnetic.

- Happens only below Critical temp.
- The Magnetic field B=0 inside the Superconductor.
- The temp of material is used than the Resistivity also used the partical temp is called Gitical temp. the Resistivity Sudden changes to zero, So it posses zero Resistivity below the Gitical temp. the temp of the material used above the Gitical temp. and act as Normal Conductor & it is also called Gitical temperature.

9.



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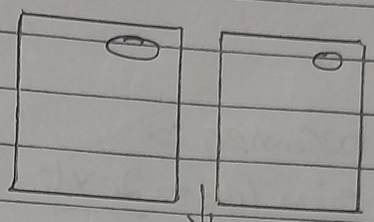
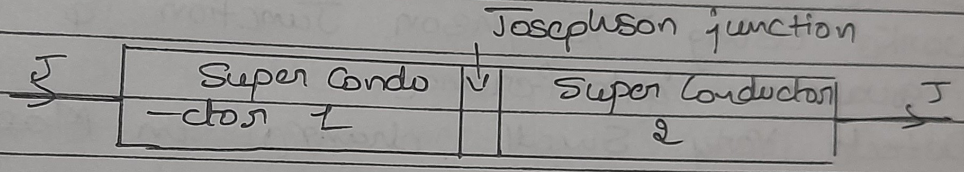
5. Elucide SQUIDS and Josephson's Effect.

=> It Consists of two superConductions seperated by using thin insulting material of thickness of 1nm - 2nm.



Insulting Material

1). DC Josephson Effect:



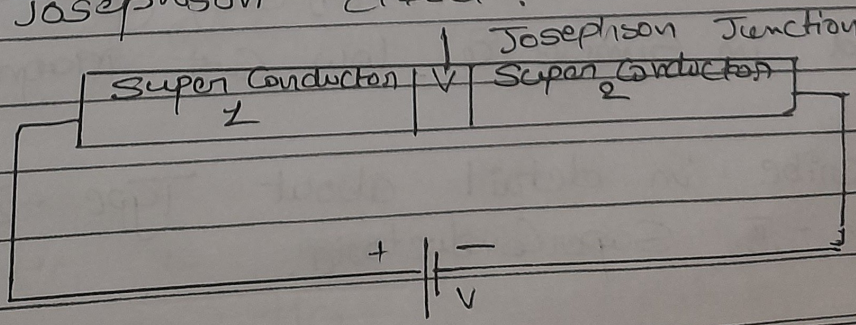
- when no Voltage is applied across the junction a Const supercurrent flows.
- This is due to quantum tunneling of Cooper pairs.
- formula: $I = I_c \sin(\phi)$

  I_c = Critical Current

  I = Super Current

  ϕ = phase difference

2). AC Josephson Effect:



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- When a voltage V is applied across the junction, the current oscillates with frequency
- Frequency of oscillation:
  $f = \frac{2eV}{h}$
- The current becomes:
  $I(t) = I_c \sin\left(\phi_0 + \frac{2eVt}{h}\right)$

⇒ SQUIDS:

- It is a very sensitive MF detector
- works using Josephson Junction & quantum interference
- Detects very small changes in Magnetic flux.

1). DC SQUIDS:

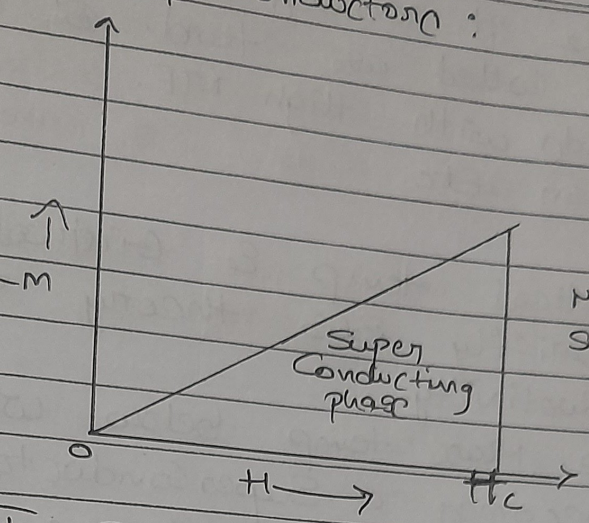
- More sensitive than RF SQUIDS:
- Made of two JJ in a superconductor loop
- operates with direct current
- output voltage varies with MF.
  $V \propto \sin(\pi\phi/\phi_0)$.

2). RF SQUIDS:

- Less sensitive than DC SQIDS
- Has one JJ in the SC loop
- works with radio frequency (RF) signal.
- used in simple & low cost Magnetic Measu

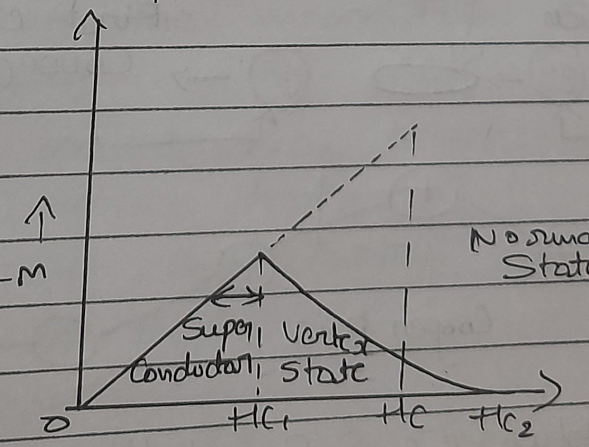
6). Describe in detail about Type - I and Type - II Superconductors.

⇒ Type I SuperConductors :



> Type I SuperConductor there are one Gitical field
> blw 0 to H_{C_1}, the Magnetization ↑ing
> Beyond H_{C_1} it becomes normal Conductors
> It is also called as Soft SuperConductor Bcc it can't with high MF.
Eg: Hg, Tin, Pb etc.

2). Type II SuperConductors :



> There are two CF H_{C_1} & H_{C_2}
> 0 to H_{C_1} ~~to~~ ~~H_{C_2}~~ Magnetization ↑ing &
blw H_{C_1} to H_{C_2} Magnetization ↓ing Called vertex ~~state~~ state

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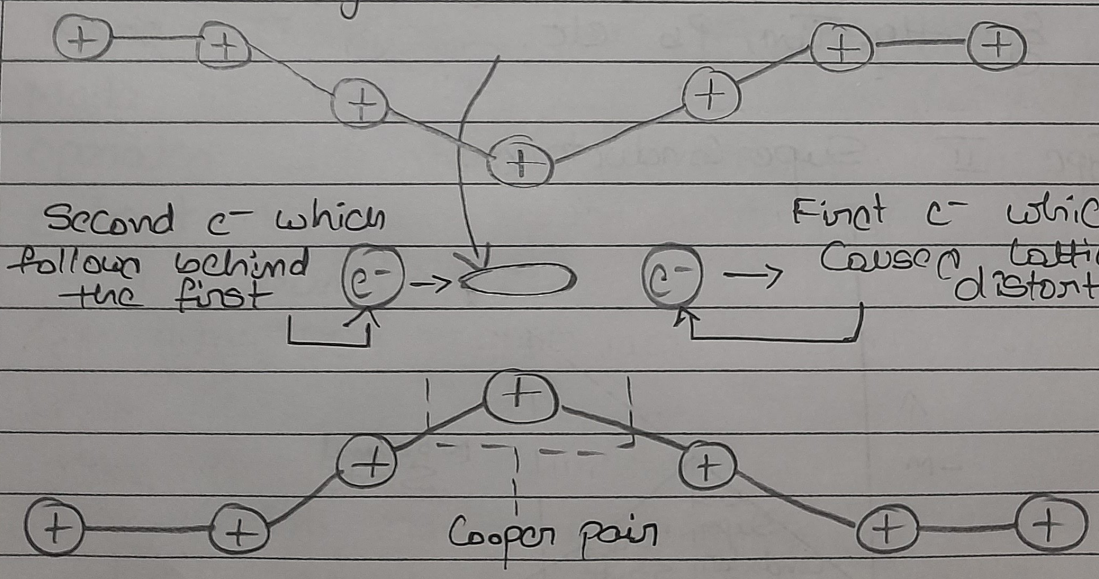
> Beyond H_{C_2} it becomes NC
> It is also called as hard SC because it stands with High MF.
Eg:- alloys, NbSn Etc.

7. Define Critical temp & Critical MF. Explain briefly BCS theory of Superconductivity.

⇒ It is the max temp below which a material becomes a Superconductor Above Tc, the Material behaves like a normal Conductor Called Critical temp

→ It is the max MF that a Superconductor Can withstand & still remain in the Superconducting State. Called Critical MF.

→ BCS Theory:



> Attraction blw e⁻s & lattice point it gives lattice distortion.
> e⁻ lattice & e⁻ interaction this lowers the Energy of both e⁻.

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> The both e⁻ had opposite Spin & momentum.
The attraction force become larger than Columbic repulsion.
> This two e⁻s will then become Copper pair.

8. Elucidate the importance of Size & Scale, weight & Strength in animation.

=> Size :

> Determines the Compatibility of ammunition with Specific weapons
> The Size of ammunition affects how it fits in to weapons & how it is stored or transported.

-> Scale :

> Affects range, accuracy, & penetration power
> Smaller rounds are Easier to Carry in large No ; Larger ones may deliver more damage but are heavier.

-> Weight :

> Influences the momentum & KE of the Projectile
> Heavier bullets may travel Slower but have more impact force
> Lightweight ammunition improves mobility & Speed in operations.

-> Strength :

> The Strength of the Casing Ensure the ammunition Can withstand high pressure during ~~it~~ firing.
> Strong materials reduce the risk of misfiring.

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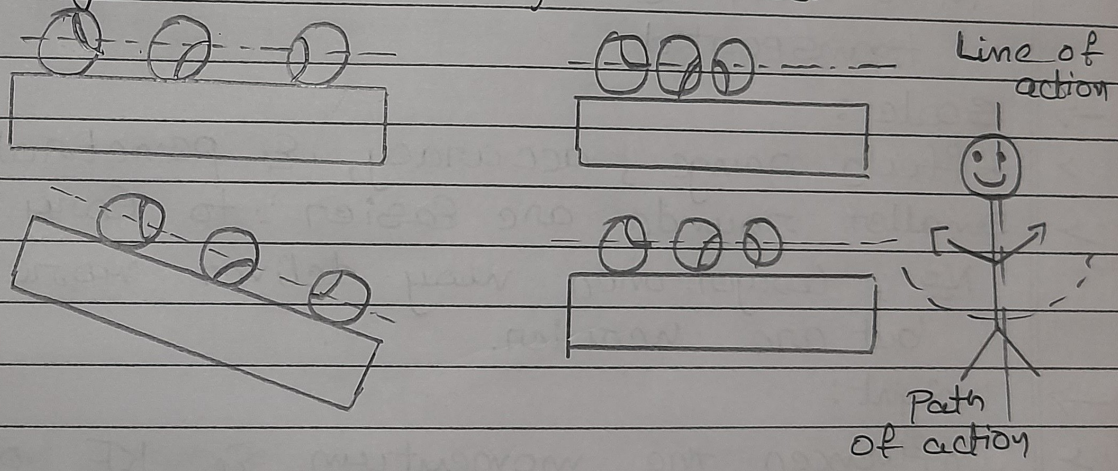
9. Discuss the timing in Linear Motion, Uniform Motion, Ease in, ease out in animation

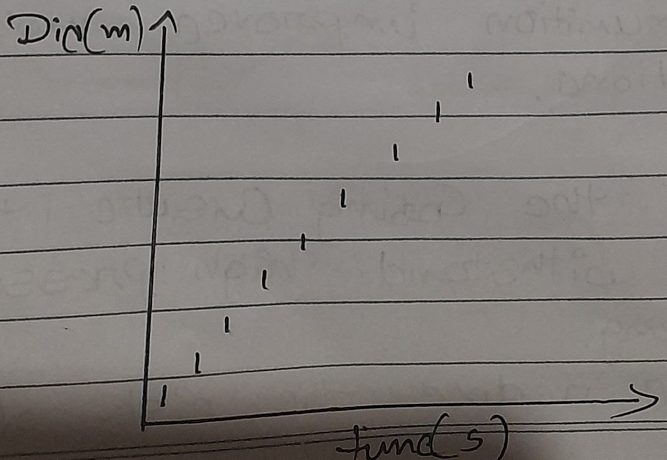
⇒ Timing animation refers to the duration of an action.

- In animation, timing of action consist of placing objects in a particular locations at specific frame.
- Individual drawings core a line of action.
- Path of action which indicates the path along which the object moves.



→ Linear Motion:

Motion that occurs in a straight line is called LM.



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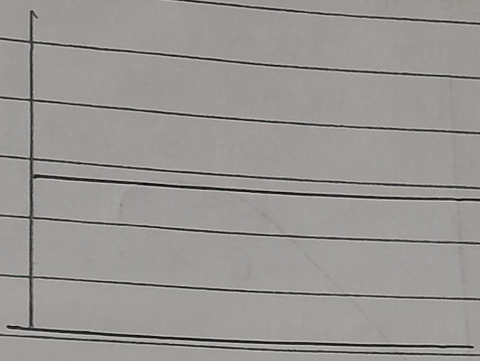
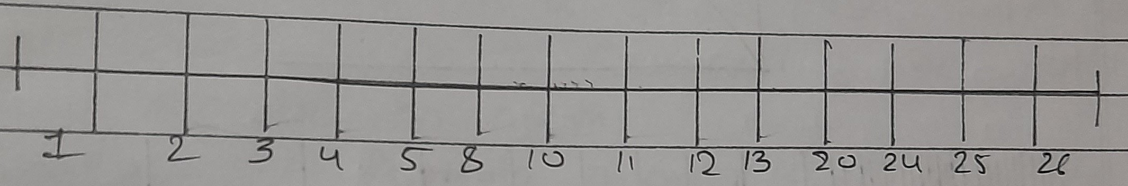
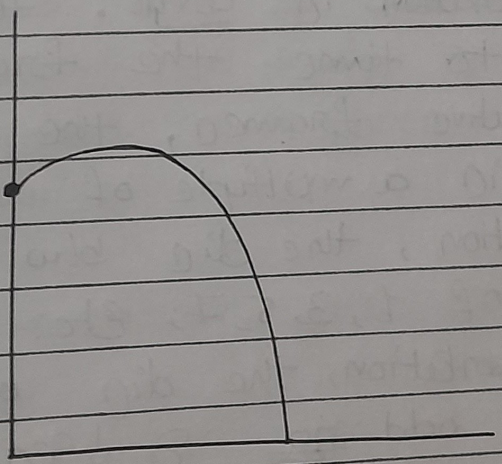
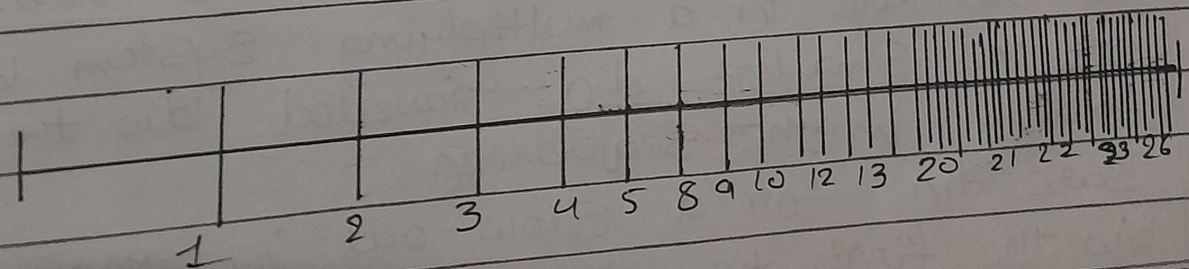
→ Uniform Motion: An object moves Equal dis in Equal time intervals. its Speed remains Constant.





→ Easy in:
The motion starts slowly & than gradually speeds up. It is used for smooth starting.





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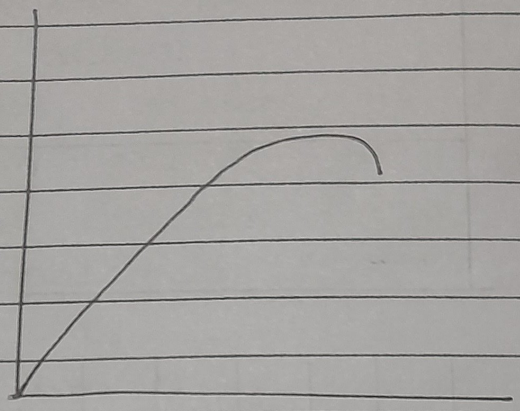
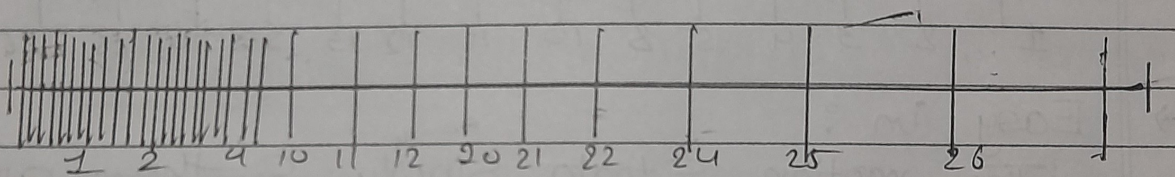
→ Easy out:
The motion starts fast & then slows down smoothly. It is used for gentle stopping.





10). Illustrate the odd rule & odd rule Multiplier with a suitable Eg:

=> When acceleration is Const. the odd rule is used to time the frames

- Blw Consecutive frames, the dis moved by the object is a multiple of an odd no
- for acceleration, the dis blw frames ↑ses by multiples of 1, 3, 5, 7, Etc.
- for deceleration, the dis of Multiples start at a higher odd no & ↓ses. like 7, 5, 3, 1.
- The odd Rule is a multiplying System based on the Smallest dis travelled blw two frames in the sequences
- Base dis, for a Slow-out is the dis blw the first two frames & for a Slow in

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the dis b/w the last two frames is called base distance

> Odd Rule Multiplier can be used to calculate the dis from the first frame to the current frame & use these dis to place the object on specific frame

> odd Multipliers for consecutive frame :- ((Frame# - 1) x 2-1).

Eg:- Suppose there is a jump push with const acceleration over 5 frames, & the total dis travelled is 0.4 m.

Using the formula above, we find the base distance.

Base distance = 0.4m / (5-1)² = 0.4 m / 16 = 0.025 m.

11). Describe jumping, parts of jump & jump magnifi-cation

=> Jumping is a type of sudden motion where an object or body quickly moves upward or forward due to a force or impulse, usually against gravity.

-> Parts of Jump:

> Crouch = A squatting pose taken as preparation for jumping.

> Take off = Character pushes up fast & straightness legs with feet still on the ground.

> In the air = Both the character's feet are off the ground.

> Landing = Character touches the ground &

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bends knees to return to a Crouch.
The dis from the character CG when her feet hit to the ground

- Push height: The dis blw CG in Crouch positions to CG of the off position
- Jump height: The dis blw CG in take-off position to CG of position at air
- Stop Height: The dis blw CG in Landing Position to CG of Crouch position during Landing.

→ Jump Magnification:

- Jump magnification occurs when a Vibration System Experience amplified motion due to resonance
- Jump Magnification is in fact an Exact ratio that tells one how much the character has to accelerate against gravity to get in to the air
- Push time: The no of frames required to move from Crouch position to Take off posn.
- Jump time: The no of frames required to move from Take off Position to in air Position.
- Stop time: The No of frames required to move from in air position to landing position

$$JM = \frac{\text{Jump Time}}{\text{Push Time}} = \frac{\text{Jump Height}}{\text{Push Height}} = \frac{\text{Push Acceleration}}{\text{Jump Acceleration}}$$

$$= \frac{\text{Push Acceleration}}{\text{Gravitational Acceleration.}}$$

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12. Distinguish b/w descriptive & inferntial Statistics.

⇒

Descriptive Statistics	Inferntial Statistics
➢ Summarizes & Presents data	➢ Draws Conclusions / Predictions from data
➢ Deals with mean, median, mode, etc	➢ Uses probability, hypothesis testing, etc
➢ Applies to Collected data only	➢ Applies results to larger population
Eg:- Average Speed of 10 measured objects	Eg:- Predicting average Speed of all objects

13. Elucidate Poisson distribution with an Eg:

⇒ Poisson distribution is a probability distribution that shows the likelihood of a no. of rare Events occurring in a fixed interval of time or Space, when the events happen independently & the average rate is Constant.

formula: $P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$

where,

$P(x)$ = Probability of x events

λ = average no. of Events

e = Euler's no

x = actual no. of occurrence

$x!$ = factorial of x

→ Suppose on average 3 Cosmic rays hit a detector Per minute.

What is the probability that Exactly 2

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Cosmic rays hit the detector in one minute?

Given, $\lambda = 3$, $x = 2$

$$P(2) = \frac{e^{-3} \cdot 3^2}{2!} = \frac{e^{-3} \cdot 9}{2}$$

Using $e^{-3} \approx 0.0498$:

$$P(2) \approx \frac{0.0498 \cdot 9}{2} = \frac{0.4482}{2} = 0.2241$$

So, the probability is = 22.41 %.

14. Slowing in object in an animation has a first frame distance 0.5 m & the first slow in frame 0.35 m. Calculate the base dis & the no of frames of frames in sequence.

=> Given;

First frame = 0.5 m

Next frame = 0.35 m

Using odd no rule

$(2n-1)d = 0.5$

$(2n-3)d = 0.35$]

Subtracting:

$2d = 0.15$ and $= 0.075$ m

Now, $(2n-1)(0.075) = 0.5$ â $2n-1 = \frac{0.5}{0.075} = 6.67$

$\Rightarrow n$â $\frac{7.67}{2} = 3.83$ â 4 frames.

→ Base distance (d or E_0) = 0.075 m

→ number of frame = *